LoDA-SA: Low-Frequency Distance-Aware Sparse Attention

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***Abstract*—A finite candidate budget must leave useful history accessible across positional regions without making selection more expensive than the attention it removes. We study LoDA-SA, Low-Frequency Distance-Aware Sparse Attention, using a two-level page-to-token cascade. LoDA-SA first specifies candidate access by distance: n_l defines a near region and n_b reserves its L1 pages, while far pages use the remaining page budget. A shared 32-coordinate low-frequency RoPE view narrows both stages' scoring dot products. The two page pools and nonaligned protection-boundary pages feed one L2 token competition, followed by original-dimensional attention. Four sink tokens and a recent window are protected separately. Qwen3 diagnostics motivate examining regional access but do not establish hybrid scoring superiority: the best tested fixed-cap hybrid has mean missed mass .135 versus .121 for global extrema. Historical task comparisons used different token allocations, and separate timings show slower sparse decoding; neither validates current LoDA-SA. Matched quality and preparation-inclusive complete-request measurements remain necessary.**

***Index Terms*—sparse attention, shared representation, regional retrieval, long-context inference**

I. Introduction

Long-context decoding must retain access to useful history under a finite attention budget. Tokens at different distances compete for this budget, and global ranking alone does not reserve candidate access for a positional region. Distance therefore offers a way to organize selection, without assuming that importance decreases monotonically with distance or that near tokens are always preferable. A selector must decide which regions can supply candidates and then rank those candidates cheaply enough to justify restricting final attention.

Cascades such as AsyncTLS shortlist pages using inexpensive summaries before token refinement [4]. Mean pooling and sign-aware extrema, summaries our selector also uses, do not rank pages by their total attention weight. Following ClusterKV's token/page perspective [8], Figure 1(a) shows a single Qwen3 query where fewer missed mass-ranked reference pages coincide with more missed reference-page mass. Unequal regional allocations and realized token counts preclude a controlled comparison, but motivate examining how scoring and budgets jointly shape candidate access.

Across 128 correlated observations from four sequences, the two global selectors have different regional missed-mass profiles (Fig. 1(b)). Mean missed mass, normalized by each query's total mass-ranked reference mass, is .084 near and .170 far for block averaging, versus .079 and .042 for sign-aware extrema. Extrema have lower mean misses in both regions; these diagnostics do not establish regional specialization or near-region equivalence. With a fixed 2048-token cap, the best of 23 tested near-avg/far-extrema splits has mean missed mass .135, versus .121 for global extrema. This bounded diagnostic scan establishes neither a hybrid advantage nor a task-quality benefit.

We organize this design around LoDA-SA: Low-Frequency Distance-Aware Sparse Attention. Its distance-aware policy specifies near extent n_l and near-page budget n_b; far pages receive the remaining L1 budget. This controls regional candidate access rather than a final near/far token share. Both stages then use one fixed set of 32 coordinates comprising 16 complete low-frequency RoPE pairs, and L2 pools admitted tokens for one TopK before original-dimensional Q/K/V attention. In Figure 1(c), these pairs retain .84 of full-dimensional reference mid-TopK mass on average, versus .51 for random coordinates and .24 for high-frequency pairs; random32 is unpaired. These diagnostics do not establish task-quality preservation. The description begins with the distance policy; execution prepares the shared representation before regional selection. Ordinary L2 still scores full history before masking, and the historical optional gate remains disabled.

LoDA-SA studies the combination of explicit distance-region page budgets, a shared low-frequency scoring view and unified token competition within an inherited two-level paradigm. Its contribution is the mechanism specification and bounded evidence for this combination, not the first page-to-token cascade or a verified quality–latency improvement. Historical budget-coupling analysis distinguishes page admission from final token coverage; historical task comparisons and slower sparse decoding constrain the conclusions. Matched current-method quality and complete-request cost, including preparation, maintenance and selection, remain open.

II. Related Work

Quest supplies sign-aware page-extremum bounds, SparQ selects key coordinates from queries, and HISA prunes the attention set hierarchically [1], [2], [3]. AsyncTLS explicitly follows Quest for coarse block selection and applies Double Sparsity for fine token selection; its cascade also includes quantization, per-query normalization with GQA probability aggregation and original-KV attention [4]. Earlier Twilight uses a select-then-prune architecture [14]. LoDA-SA inherits this broad hierarchical-selection paradigm and studies explicit distance-region page access with complete low-frequency RoPE pairs shared across both stages and unified L2 competition. This relationship concerns algorithmic ideas, not verified direct code inheritance.

The selector's own reads also matter. Token-Picker uses conservative probability estimates from partial key-bit chunks to stop further key-chunk and value transfers for pruned tokens [13]; our selector instead ranks candidates with a fixed coordinate subset under page and token budgets, keeping original-dimensional KV. Our bf16 configuration stores floating-point scoring keys; AsyncTLS uses packed INT4 token-level indices and FFD scans 2-bit key thumbnails [4], [5]. Compact coordinates and regional quotas control scoring width and candidate access. KV compression and placement methods address complementary storage and movement constraints [6], [7], [8]; a smaller scoring view retains original-KV capacity and alone establishes no complete-request latency benefit.

III. LoDA-SA: Distance-Aware Selection in a Shared Subspace

LoDA-SA separates three decisions: which distance regions can admit pages, how to score them with a shared compact view, and which pooled tokens enter final attention. The distance policy uses n_l for near extent and n_b for its page budget, leaving $K_1 - n_b$ far pages. The implementation prepares the shared 32-coordinate representation before applying regional L1 selection (Fig. 2); this execution order is distinct from our distance-first explanation. Selection runs only during decoding, with dense prefill. Reference L2 scores full history, masks by the combined page candidates and performs one middle-token

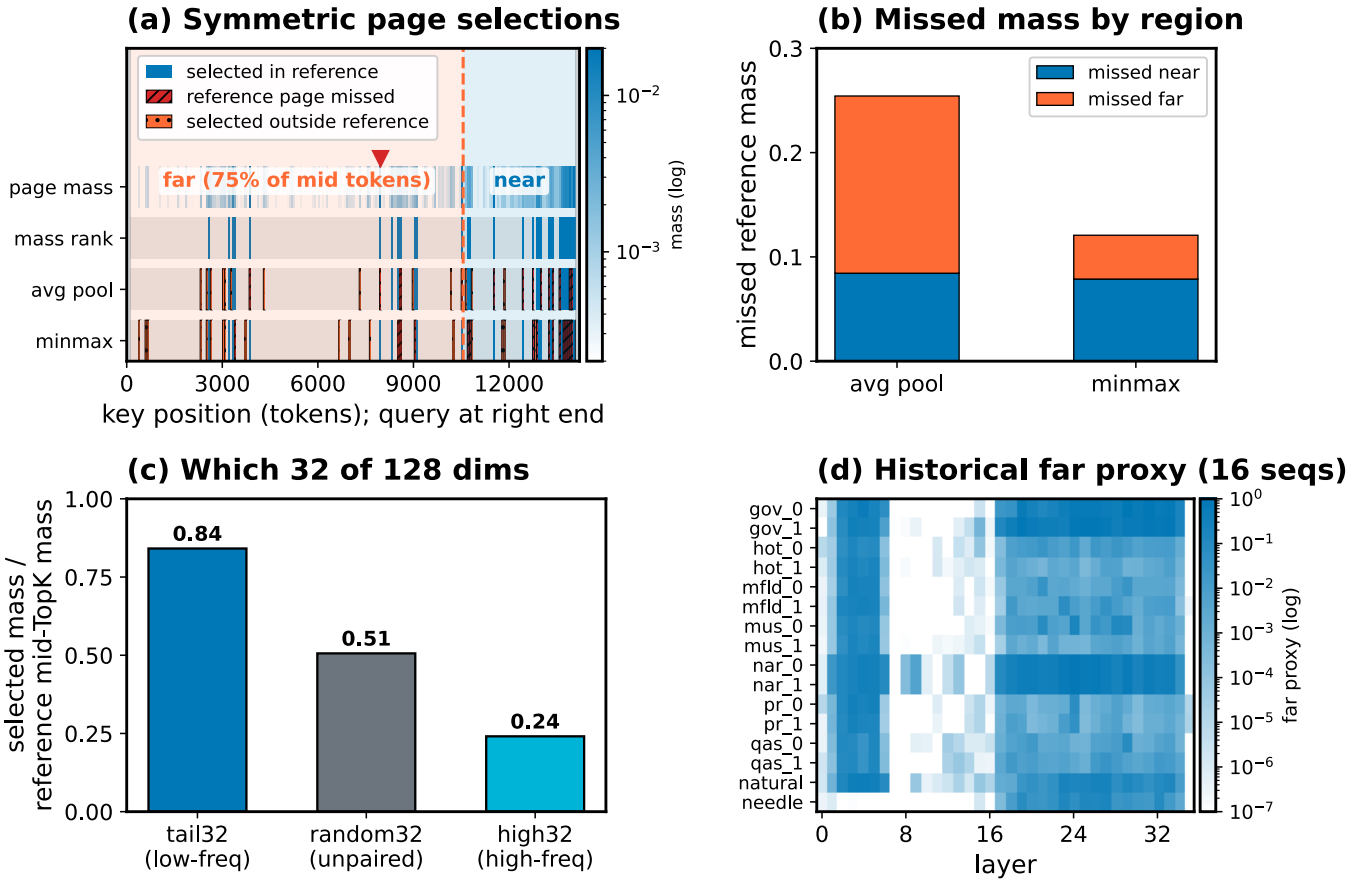


Fig. 1. Motivating diagnostics for LoDA-SA; Qwen3-8B, 128/128 prefix/recent protection in (a)–(c). (a) MuSiQue sample 1, layer 24, query 14212. The top strip shows summed attention per page, not a selection. Below: stored mass-rank, avg and minmax selections, 32 pages each, using 1989/1989/2048 tokens under a 2048 cap. Avg/minmax miss 13/17 reference pages holding 59.65%/26.46% of reference-page mass. Blue: selected reference pages; red hatching: missed reference; orange dots: selected outside the reference. Mass-rank is a whole-page comparison reference, not an optimal oracle, exact token TopK or DSA. Avg and minmax share 32 coordinates and head aggregation; the reference uses full-dimensional attention weights. Different regional allocations and realized counts preclude inferring near/far specialization or task quality from these misses. (b) Mean regional shares of missed reference mass: for each query, missed reference mass in each region is divided by total mass-ranked reference mass, then averaged across 128 correlated observations from four sequences. Extrema miss less in both regions; this does not establish regional specialization. (c) Selected mass relative to full-dimensional reference mid-TopK mass; random32 does not preserve RoPE pairs. (d) One 16-by-36 historical single-query proxy matrix gives 24/36 layer medians below .01 and 12–32 low-proxy layers per sequence. It uses the last sampled query, group-logit summation before softmax and far interval $[64, t-2048)$ (S6), unlike the historical optional gate statistic. These are diagnostic selections and profiles, not task-quality or decode-safety results.

TopK. Winners and protected indices define the original-dimensional attention support. Gating and candidate gathering are disabled in this reference configuration; archived runs do not validate current LoDA-SA.

A. Distance regions and candidate budgets

In decoding, t indexes the current token from zero across the full sequence including prefill. Let s and w denote the protected sink and recent-window lengths, with protected set $\mathcal{P}_t = \{i : 0 \le i \le t,\ i < s \text{ or } i > t-w\}$ and unprotected middle $\mathcal{M}_t = \{0, \dots, t\} \setminus \mathcal{P}_t$; overlapping protection counts once. The requested near length is n_l tokens. The implementation uses at least one page, $L = \max(B, n_l)$, then floors the near/far cut to a page boundary; the resulting near region ends at the protected window. Far is the remaining middle history. Near excludes the protected recent window.

For ordinary long histories with disjoint protection and total cap $K \ge |\mathcal{P}_t|$, define $K_{\text{mid}} = K - |\mathcal{P}_t|$. LoDA-SA uses n_l and the near-page budget n_b, with far-page budget $K_1 - n_b$; page quotas are clipped to available pages. Selected near and far pages, together with protection-boundary bypass pages, form one L2 candidate pool. A single TopK uses K_{mid} slots across that pool. There is no prescribed near/far token split: regional representation in the final set follows the scores, not a token quota. With enough positive-probability candidates, L2 cannot recover unprotected tokens from L1-rejected pages. S7 details scarce-support and masked zero-tie exceptions.

With disjoint protection, $K = 1024$, $s = 4$ and $w = 128$ leave a nominal 892 shared middle slots per KV head. A partial sink page can select protected positions again; the final union deduplicates them, so the number of distinct unprotected winners can be smaller. Page-level eligibility is therefore distinct from both final inclusion and exact budget utilization.

B. Shared low-frequency coordinates

Both regions and both selection stages use the same coordinate map within each layer and KV head, while each query head is scored separately. For the Qwen3-8B tail-coordinate configuration, we select post-RoPE coordinates $J = \{48, \dots, 63, 112, \dots, 127\}$ from each 128-dimensional query and key. The indices retain 16 complete low-frequency split-half RoPE pairs (Supplementary Fig. S1); this statement does not describe generic full-width, static-pair, projection

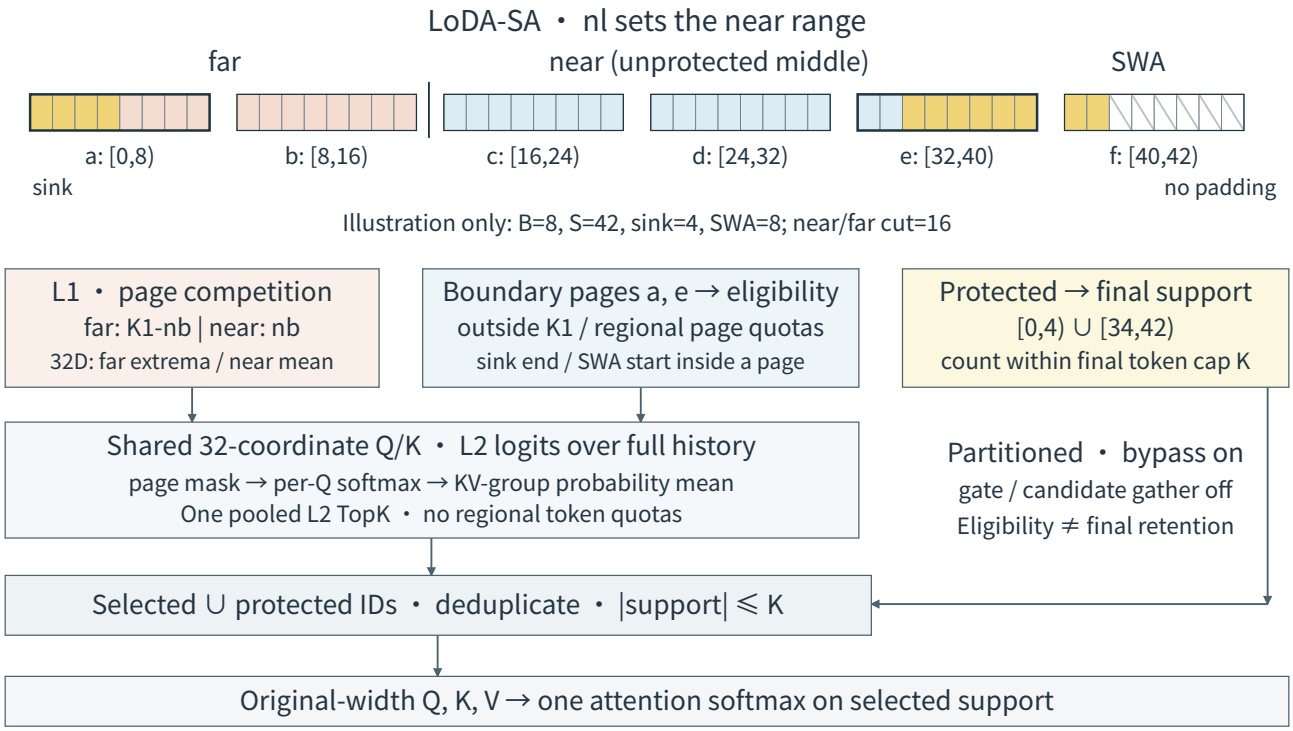


Fig. 2. LoDA-SA partitioned reference architecture: boundary bypass enabled, gating and candidate gathering disabled. A shared coordinate view feeds regional L1 page scoring and ordinary full-history L2 scoring. One chronological page sequence is partitioned into far and near. Nonaligned protection-boundary pages join the page mask outside L1 quotas; only protected tokens bypass final selection. Page masks limit eligibility; per-query softmax and KV-group probability averaging precede one pooled middle-token TopK. Protected positions join deduplicated IDs within K. Near excludes the protected recent window. IDs define support over original-width K/V; physical gathering is backend-dependent. Arrays are schematic.

or MLA configurations. With coordinate-selection matrix $E_J \in \mathbb{R}^{128\times 32}$, the scoring vectors are $z_q = E_J^\top q$ and $z_i = E_J^\top k_i$. The retained pairs' slow positional rotation motivates this selection, although it does not guarantee preservation of important tokens. The compact implementation uses width-32 scoring tensors, whereas the historical quality runs used 32 active coordinates in width-128 L2 tensors (S7).

L1 aggregates selected key coordinates into page summaries for coarse ranking; L2 refines candidates with per-token scores in the same coordinate view. The two scoring views occupy separate buffers. Original KV remain full width. Compact scoring narrows each dot product fourfold, but view preparation and maintenance add work without compressing the original KV cache (S7).

C. Regional page scoring and boundary eligibility

Regional page caps and one shared token budget constrain the cascade (Supplementary Fig. S2). Near pages use query–mean-key dot products to summarize page-level relevance; far pages use sign-aware extrema to bound shared-space token scores, a heuristic assignment. Summaries cover full physical pages, including protected positions (S7). In the partitioned path, a page containing a non-page-aligned sink end or recent-window start bypasses L1 ranking: it is removed from the competition pools and appended without consuming their page quotas. Its unprotected tokens still compete in L2; only protected tokens are forced into final support. The near/far cut is page-aligned and adds no bypass. For a nonempty page b, let $m_{b,j} = \min_{i\in b} z_{i,j}$ and $M_{b,j} = \max_{i\in b} z_{i,j}$. Quest's construction [1] gives

$$U_b(z_q) = \sum_j (z_{q,j}^+ M_{b,j} + z_{q,j}^- m_{b,j}), \tag{1}$$

where $z^+ = \max(z, 0)$ and $z^- = \min(z, 0)$. Each signed coordinate contribution is bounded by its chosen extremum, so $z_q^\top z_i \le U_b(z_q)$ for every shared-space key in b. This does not bound original-dimensional logits. Extrema may come from different tokens, making the bound loose enough to misrank pages and miss important tokens. Average-key scores lack this per-token bound.

D. GQA selection and original-dimensional attention

GQA assigns four of Qwen3's 32 query heads to each of its eight KV heads. L1 averages their page scores before regional TopK. Ordinary L2 computes each query's full-history logits, applies the shared page mask, then softmax. Group-averaged probabilities determine one TopK across the pooled middle tokens. L2 uses scale $1/\sqrt{128}$ before softmax even with 32 scoring coordinates. Because softmax precedes group averaging, this rule generally differs from applying softmax to group-averaged logits. A smaller L1 shortlist restricts eligibility without reducing ordinary L2's full-history dot products.

For KV group h, let $\mathcal{S}_{t,h}$ be the deduplicated union of protected positions and pooled winners, with $|\mathcal{S}_{t,h}| \le K$. Each query g uses this support for its own full-dimensional attention ($d = 128$):

$$o_{h,g} = \sum_{i\in\mathcal{S}_{t,h}} \frac{e^{q_{h,g}^\top k_{i,h}/\sqrt{d}}}{\sum_{j\in\mathcal{S}_{t,h}} e^{q_{h,g}^\top k_{j,h}/\sqrt{d}}} v_{i,h}. \tag{2}$$

Each query applies one softmax over the merged indices, avoiding the addition of separately normalized regional outputs [9]. Full-dimensional retained logits still use a restricted denominator, which can change the output. These indices define a shared logical support.

TABLE I
Compact design parameters and execution policy. Explicit n_l, n_b values remain configuration parameters, not verified measurements here. K includes unique protected tokens.

Parameter	Setting
Final token cap K	1024
Page size; L1 competition cap K_1	64; 128
Near page budget (far)	n_b ($K_1 - n_b$)
L2 middle-token selection	one pooled TopK
Near extent	n_l tokens; page-aligned cut
L1/L2 active coordinates	32/32
Prefix / recent protection	4/128
Prefill; decode layers	dense; all sparse

Physical KV gathering and reuse depend on backend and length; short sequences can use a dense masked path.

E. Scope of optional far gating

The current n_l/n_b design keeps far gating disabled. A historical extension calibrates far mass from final prefill queries after per-head causal softmax, then freezes a layer decision at sparse decoding. Its numeric-token-quota branch can refill a near-token quota, but that behavior is not the current unified L2 rule (S5). In the unified branch, removing ordinary far-page competition does not guarantee removal of all final far tokens: a sink-boundary page may still enter the pooled competition. We therefore do not claim complete far skipping or transfer historical gate results to this design. Calibration thresholds do not bound decode error, and both quality and complete-request cost require separate evaluation.

IV. Experimental Setting

We examine task quality (Q1), regional candidate coverage (Q2) and execution cost (Q3) separately. Table I specifies LoDA-SA for Qwen3-8B: dense prefill, sparse decoding, explicit n_l/n_b page selection and unified L2, with a 1024-token cap including four sink and 128 recent tokens. The current method does not tune α, β, γ or a near-token quota. No matched completed run of this configuration is established by the records below. Historical numeric-token-quota runs retain their original identities; $(8, 320)$ denotes near pages and near tokens, not (n_l, n_b). Historical LongBench uses 128/128 protection and 32 active coordinates in width-128 L2 keys, with 16-level quantize–dequantize (S7). Nominal full-dimensional page/token variants lack complete execution configurations and are not controlled single-stage ablations. Selector and decode timings come from separate studies without matched quality or cost.

Workloads and comparisons. Reported quality covers five-task LongBench [10] and 11-task RULER [11] (Qwen3-8B: 32K/64K and historical 128K), plus ported Quest/AsyncTLS/JSSA baselines. Three nominal sparse LongBench variants lack verified factorial isolation. FullKV is dense and uncapped (RULER/decode timing). Method comparisons must retain and disclose each baseline's native prefill/decode path; policy-matched adaptations need separate reporting. Budget matching aligns final unique-token caps K including protection; realized counts and head sharing affect KV traffic, so equal K alone establishes neither quality nor cost matching (S7).

Metrics. Primary quality is task-score AVG over each fixed benchmark/length set; per-task gains and losses explain AVG without a zero-loss requirement. Policy-matched controls require matched models, samples, lengths, scorers, phase policies and aligned sparse-token budgets; incomplete sets lack full-set AVG. Panel (b) reports regional missed-reference-mass shares normalized by each query's total reference mass and averaged over correlated observations; panel (c) uses full-dimensional reference mid-TopK mass. Neither selector time (synthetic single-layer) nor synchronized TPOT (model-level post-first-output intervals) measures preparation-inclusive complete-request latency. Unpaired quality and timing studies do not define a quality–latency frontier; AVG differences are descriptive, not statistical non-inferiority evidence.

V. Evaluation and Discussion

A. Q1: Does reduced scoring width preserve task quality?

On HotpotQA, Qasper, MuSiQue, GovReport and RepoBench-P [10], five-task means fall within a 0.33-point spread: shared-subspace 47.588, the nominal page variant 47.910 and the token variant 47.666 (Supplementary Fig. S4); the shared-subspace arm trails the two variants by 0.322 and 0.078. These differences describe recorded configurations, not evidence that reduced width preserves quality: the variants are not verified single-factor interventions, and the shared-subspace run uses earlier 128/128 protection. All three arms are sparse; dense FullKV is a different reference.

B. Q2: How do regional budgets constrain candidate access?

The comparisons in this subsection retain historical numeric-token-quota identities and do not validate current n_l/n_b settings. Figure 1(b) shows regional missed-mass profiles of global selectors, not a benefit from regional quotas; the fixed-cap hybrid scan also retains a negative result (.135 versus .121). Reported 8B 32K AVG rises 58.30→59.29 when near quotas move $(16, 640) \rightarrow (8, 320)$ at $K_1 = 128$, $K = 1024$, protection 4/128 (archived $\beta = .125 \rightarrow .0625$); far pages rise 112→120 and the far token budget 252→572 jointly. At archived $\beta = .03125$, $K_1 = 128$ scores 58.07 (near pages 4, far tokens 732); $K_1 = 256$ scores 58.96 (near quotas $(8, 320)$; far pages 248 versus 120 for the 59.29 arm). Both score below 59.29; $K_1 = 256$ CWE/FWE differences from that reference are $+3.9/-9.67$. Data/scorer records match these sparse arms; predictions remain unreplayed.

Historical ratio coupling. Earlier runs derived $n_p = \max(1, \text{round}(\beta K_1))$ and $n_k = \min(\lfloor \gamma n_p B \rfloor, K_{\text{mid}})$, with ties-to-even rounding. Their nominal far budget was

$$k_f = K_{\text{mid}} - \min(\lfloor \gamma n_p B \rfloor, K_{\text{mid}}). \tag{3}$$

This is a historical numeric-token-quota rule, not the current unified L2 algorithm.

Implicit coupling failures. Archived diagnostics exhibit all three reallocations of Eq. (3). Integer enumeration of the archived formulas locates the first far-zero page cap at $K_1 = 156$ for $K = 1024$ ($B = 64$, $\beta = .125$, $\gamma = .625$, 128/128 protection); far budget 8 remains at $K_1 = 155$, so capacity appears to grow while far tokens vanish. Archived task scans collapse HotpotQA from 53.6–57.3 to 37.7 exactly on zero-far-budget arms, graded by far fraction (16.4 points at .875, about 1 at .125). At the archived grid cell $(\alpha, \beta) = (.25, .125)$, free competition ties the best γ within .001 in retained-mass diagnostics (both about .852), while $\gamma = 0$ reads as far-free but zeroes the near token quota, dropping retained mass to .827. The historical 128K RULER arm computed $k_f = 0$ under its reported budgets (S6). These archived diagnostics unify the three anomalies but are not new controlled experiments.

The reported Qwen3-8B 32K AVG gap to dense FullKV is -0.09, across machines and different scoring protocols. At 64K, 8B AVGs are 48.85/48.05 for near quotas (8,320)/(16,640), versus 48.54 for a separate-run dense anchor. The sparse arms share a machine and scorer, but prediction protocols differ; the dense comparison also uses a different scorer. These descriptive comparisons establish neither causality nor statistical non-inferiority (S8).

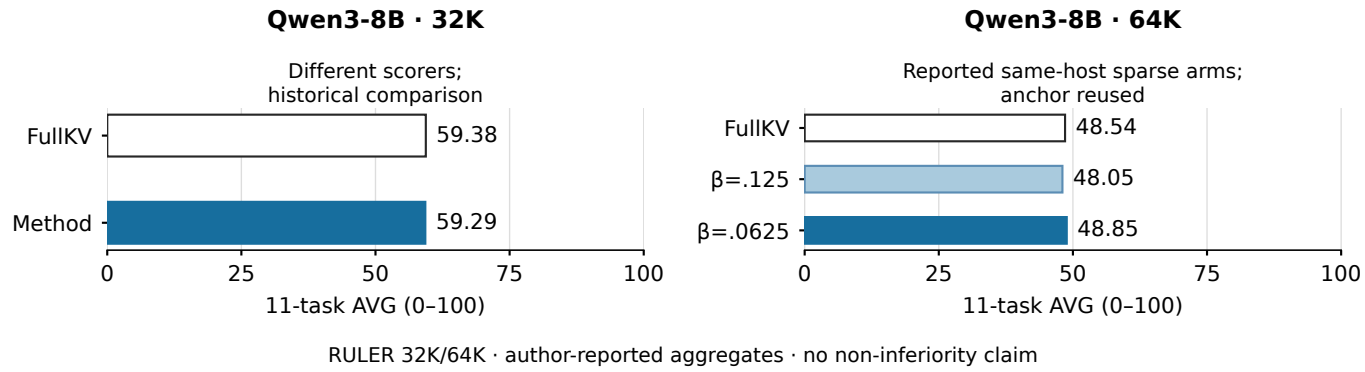


Fig. 3. Author-reported Qwen3-8B RULER 32K/64K AVG: 11 tasks, 100 examples/task/arm. FullKV is dense; sparse arms report $K = 1024$, protection 4/128. These historical numeric-token-quota arms do not evaluate the current unified L2 design. The 32K panel shows near quotas (8,320); the 64K panel includes (8,320) and (16,640). The 32K comparison spans machines with different scorers; the 64K sparse arms are reported on one machine, the FullKV anchor a separate run. Gate status and scoring dimensions are unverified. These descriptive scores establish neither non-inferiority nor matched quality–cost benefit (S8).

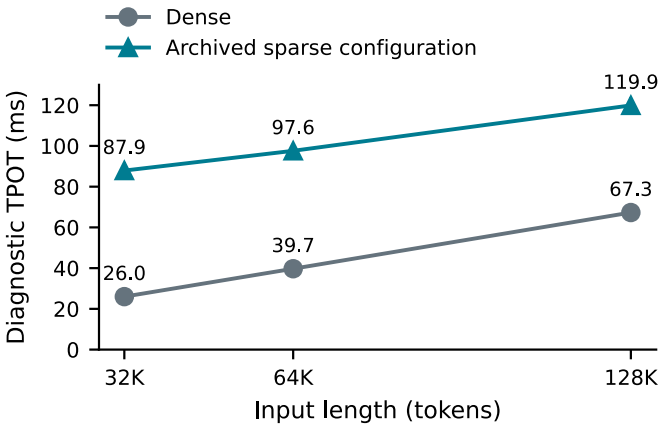


Fig. 4. Author-reported model-level decode TPOT for an archived sparse configuration and dense FullKV, Qwen3-8B with dynamic cache. Fixed 128-token output gives 127 after-first intervals, including synchronization and argmax. The archived $(.125, .375, .625)$ setting differs from the current n_l/n_b design; these medians measure no complete-request E2E and establish no matched quality–cost comparison (S6).

Ported baselines. Author-reported AVGs (nominal 512): 56.00 (ours) and 48.33 (AsyncTLS), a reported 7.67-point gap; final unique-token budgets, protection and execution phases have not been matched (S8). At $K = 1024$, 59.29 compares with JSSA's 58.7 (nominal 1024), Quest's 42.85 and FASA's 18.64; $K = 384$ scores 15.00. S8 gives provenance, CWE's 43.4-point loss and absent paired intervals.

At 128K, the (8,320)/(16,640) arms score 45.73/46.02 on one machine; 45.73 is 98.49% of the separate-run dense anchor, 46.43. Historical RULER 128K yields 23.98 (sparse) under budgets implying $k_f = 0$, execution unverified, the 22.45-point loss unexplained (S6). Five-task LongBench means are 26.09 (protection-only, 256 tokens) versus 47.91 (reused sparse, 1024); retrieval and budget both change.

C. Q3: Does compact scoring reduce execution cost?

Model-level decoding. The reported sparse configuration is slower than dense FullKV at every measured length (Fig. 4). Sparse TPOT medians are 87.9, 97.6 and 119.9 ms at 32K, 64K and 128K, compared with dense medians of 26.0, 39.7 and 67.3 ms. The relative penalty shrinks with length, but absolute sparse TPOT increases: no decode-time benefit. Timings include synchronization and argmax, use a non-compact configuration, and cannot locate causes or predict the compact implementation (S6).

Local selector cost. A separate two-round synthetic experiment (single layer, batch one, one query, 64K QKV) reports steady-state summaries of 1.3502 ms (full-dimensional page variant), 1.3598 ms (shared-subspace) and 1.4634 ms (token variant); shared-subspace runs 0.0237 ms slower in round one and 0.0046 ms faster in round two. Missing raw sequences and an unresolved sampling discrepancy prevent assessing these small differences (S6). These local timings identify no bottleneck, lack quality pairing and measure no complete-request cost.

D. Optional gate: prefill is not a decode guarantee

Low prefill attention does not establish safe decode-time skipping. An archived input–layer record pairs a prefill far-attention proxy below .05 with a decode-tail mean of .1678 over five queries: a proxy shift, not measured quality loss. Gate thresholds .05 and .10 yield five-task AVGs of 47.876 and 47.692 against the reused 47.910 reference, down .034 and .218, with three tasks declining per arm, led by HotpotQA's 1.05 and 1.26. AVG remains primary; no per-task zero-loss requirement applies. Neither these changes nor zero-containing task-bootstrap intervals establish non-inferiority; missing scoring configurations, paired predictions and matched costs leave gated quality and net savings unverified (S6).

Design implications and limitations. Two constraints recur: reduced scoring width narrows each dot product while ordinary L2 still scores the full history, and far-page quotas admit candidates without guaranteeing far-token capacity. A useful operating point reduces complete-request latency at matched quality, counting preparation, maintenance and selection once. The archived sparse configuration decodes slower than dense, selector timings establish no request-level benefit, and matched measurements for the compact design remain missing; historical gate summaries lack complete configuration identity.

VI. Conclusion

LoDA-SA organizes finite-budget sparse attention around distance-aware candidate access: n_l sets near extent, n_b reserves near L1 pages, and the remaining page budget serves far history. Shared low-frequency RoPE coordinates narrow scoring in both stages; one pooled L2 selects tokens for original-dimensional attention. Boundary pages receive eligibility, whereas sink and window tokens receive

final protection. Neither regional admission nor the low-frequency view guarantees final regional coverage or task quality. The inherited cascade is not a novelty claim by itself. The negative fixed-cap hybrid diagnostic and slower historical sparse decoding remain limitations; matched quality and complete-request cost are required to establish LoDA-SA's benefit.

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